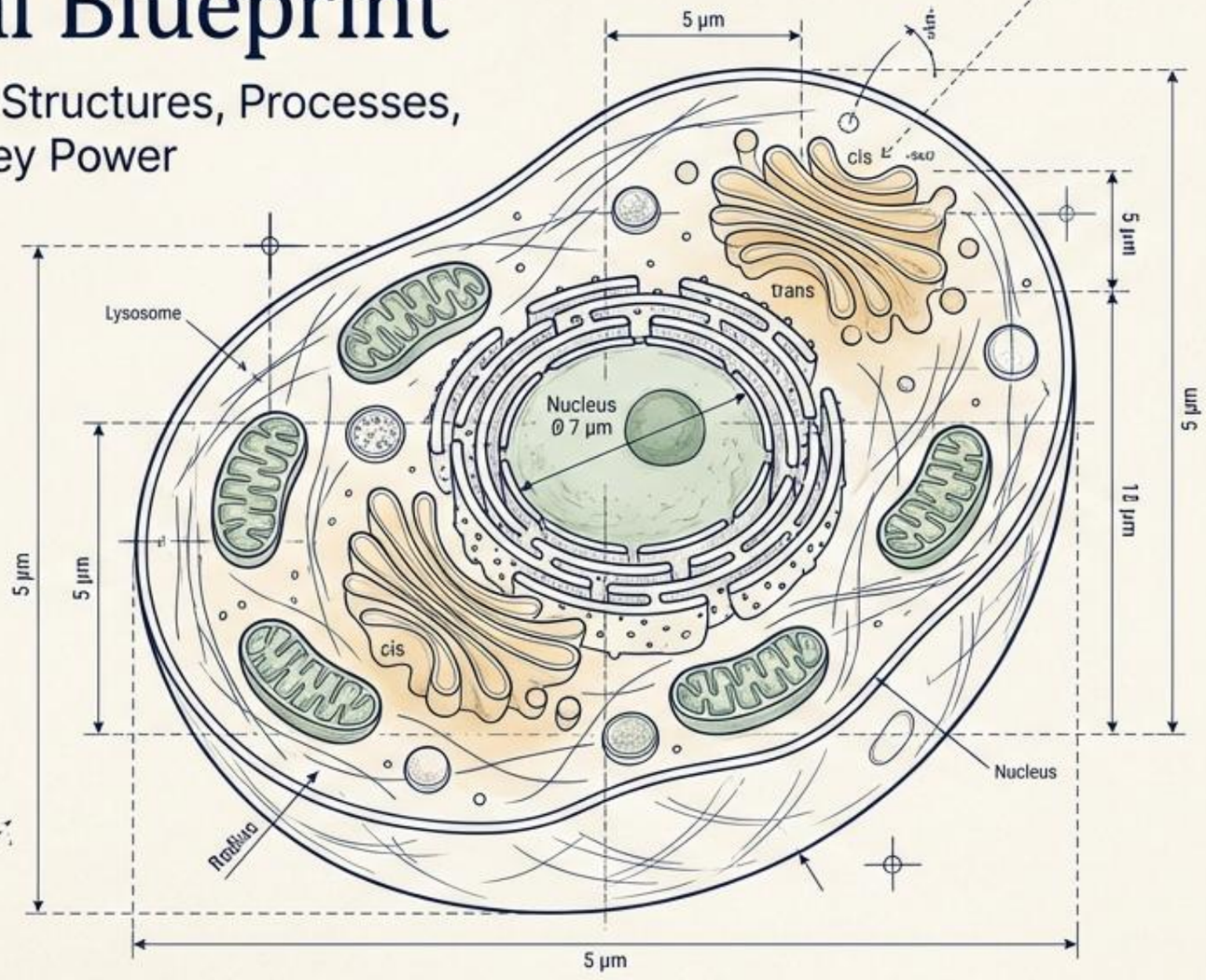
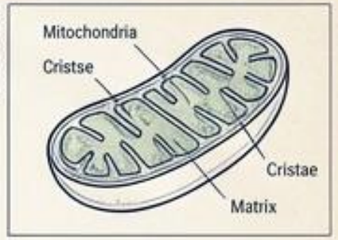
# The Architecture of Life: A Functional Blueprint

## Biology 30 Review: Cell Structures, Processes, and the Ecosystems They Power

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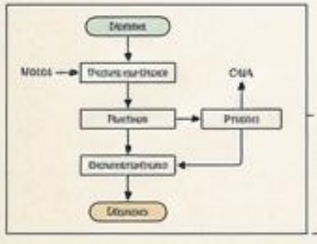

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## Session Brief

Welcome to the 20-minute guided teardown. Following this overview, you will have 30 minutes of independent study driven by the prompts embedded in this deck.

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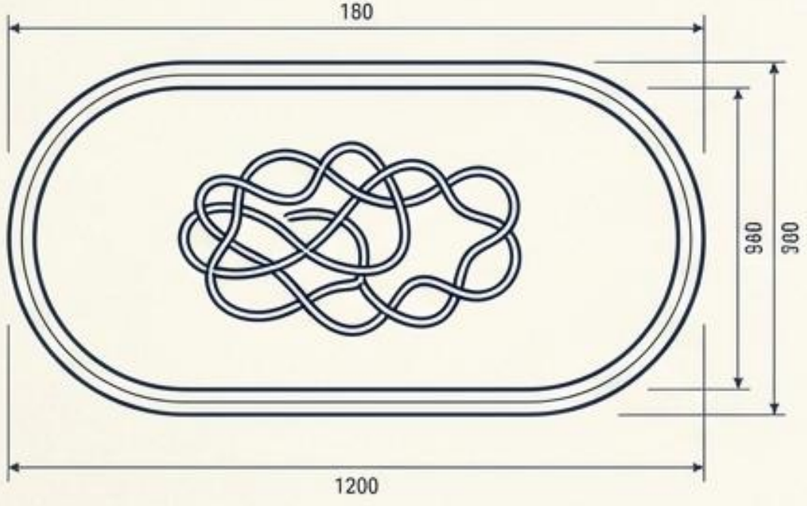
# The Two Empires: Prokaryotic vs. Eukaryotic

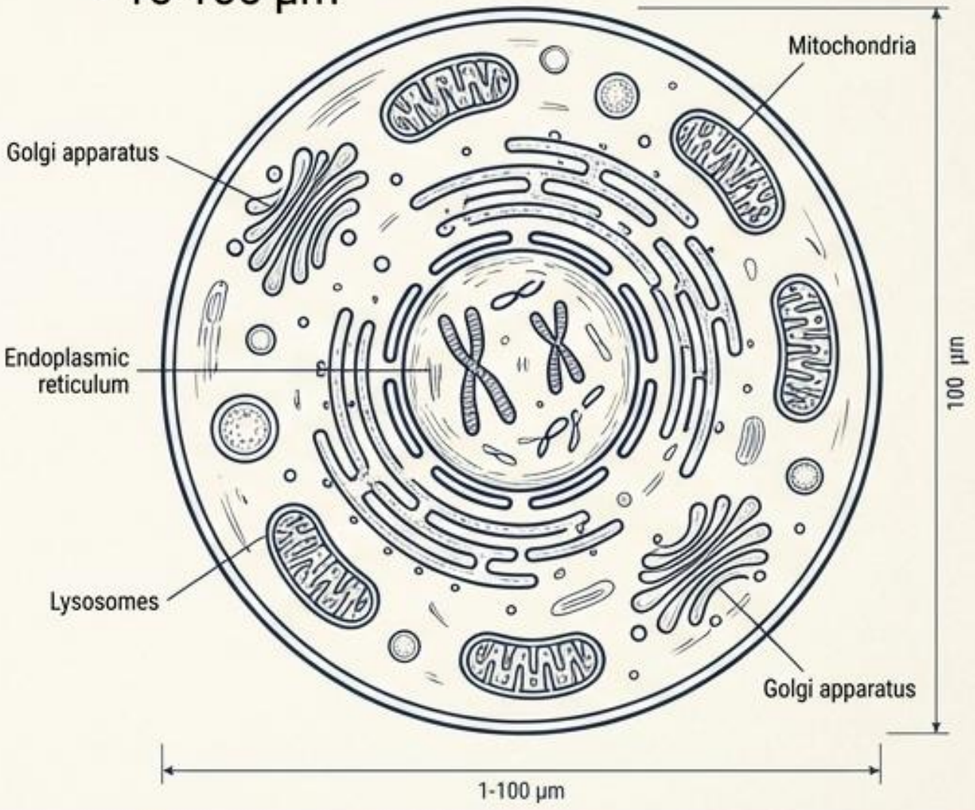
## Prokaryote (Before Nucleus)

- 1-5 µm
- No Nucleus (Free-floating Nucleoid)
- No Membrane-Bound Organelles
- Circular DNA Loop




## Eukaryote (True Nucleus)

- 10-100 µm




**Saskatchewan Context: Pulse Crops & Nitrogen Fixation**

Saskatchewan leads the world in lentils. These crops rely entirely on a symbiotic relationship with Rhizobium—soil-dwelling prokaryotes that fix atmospheric nitrogen (a feat eukaryotic plants cannot do alone).

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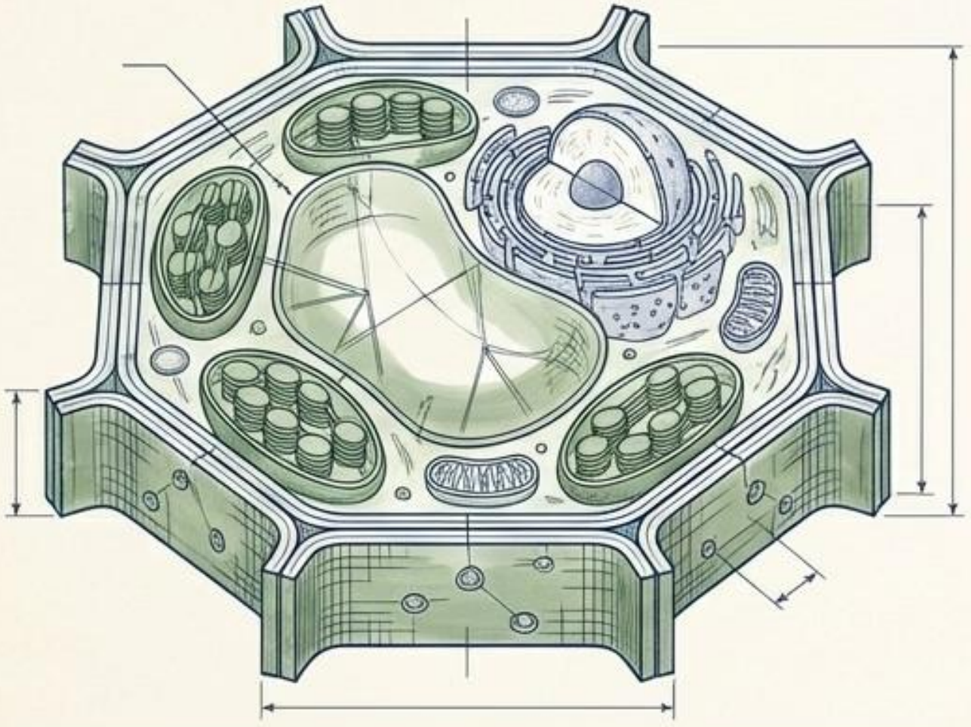
# The Eukaryotic Divide: Plant vs. Animal

**Context: The Canola Cell.**
Packed with chloroplasts to maximize short summers, and specialized oleosomes storing valuable lipids.

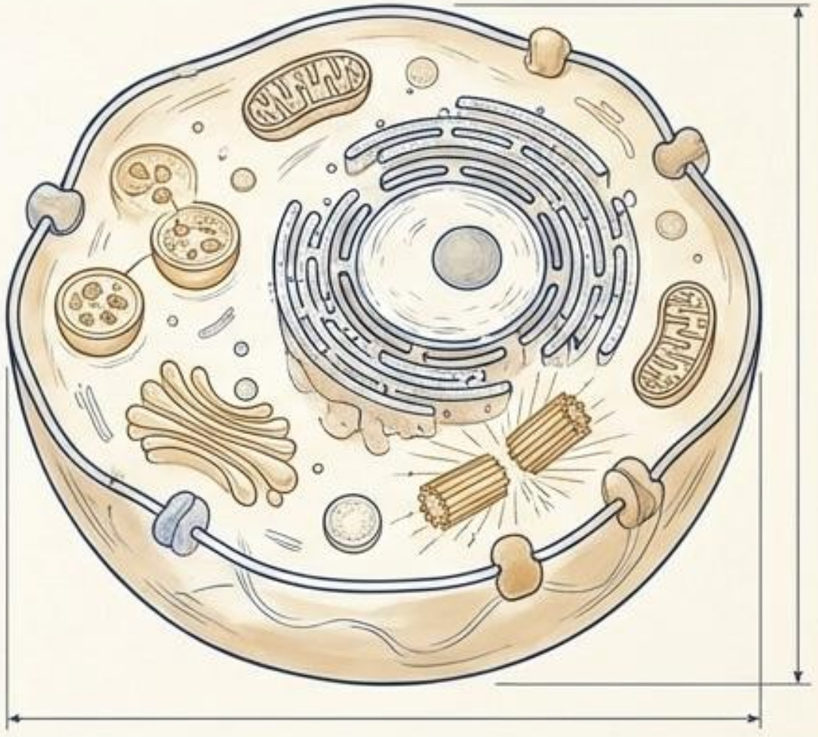
## Plant Exclusives (The Autotroph):

[✓] Cell Wall
[✓] Chloroplasts
[✓] Large Central Vacuole (structural support)



## Animal Exclusives (The Heterotroph):

[✓] Lysosomes (waste/food digestion)
[✓] Centrioles (mitosis organization)



Shared Foundation:
Cell Membrane, Nucleus, Mitochondria, ER, Golgi.

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# The Materials: Four Macromolecules

## Carbohydrates (CH2O - 1:2:1)

**Fuel & Structure**

**Monomer:** Monosaccharides (Glucose)

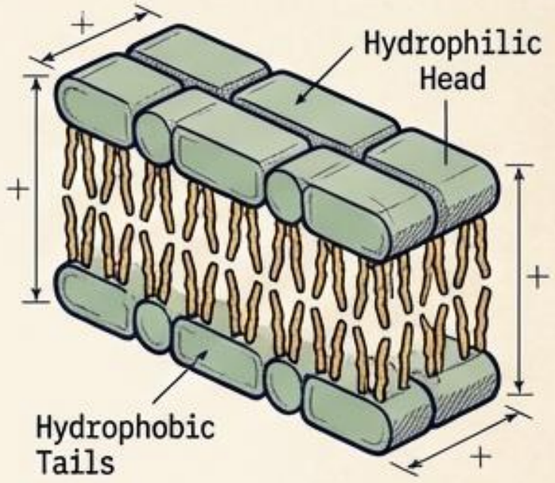
**Polymers:** Starch, Glycogen, Cellulose.

## Lipids

**The Boundary Architects**

Hydrophobic

**Key structure:** Phospholipids (Hydrophilic head + Hydrophobic tails)

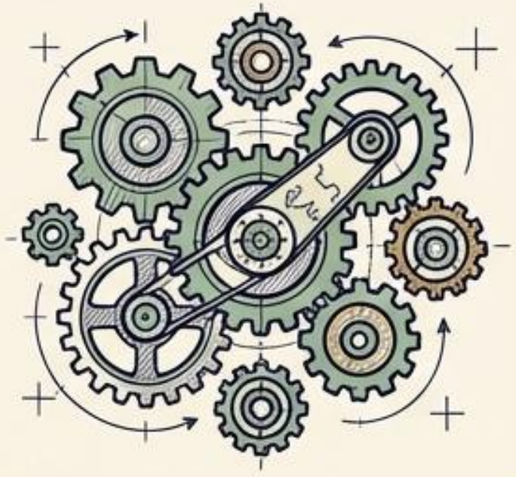



## Proteins

**The Workhorses**

**Monomer:** 20 Amino Acids linked by peptide bonds.

**Functions:** Enzymes, transport, structure.

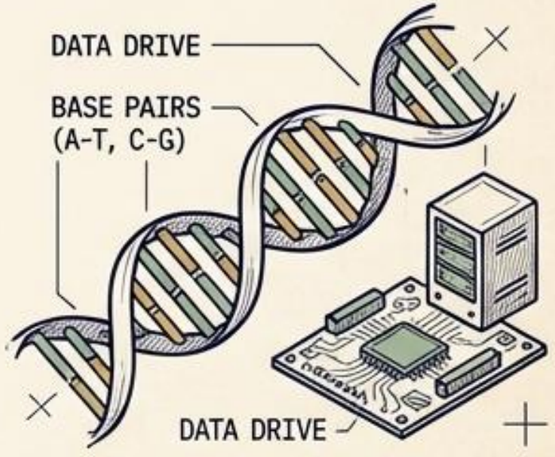


## Nucleic Acids

**Information Storage**

**DNA** (Double helix, A-T / C-G pairs)

**RNA** (Single strand, A-U / C-G pairs)




**Context:** Saskatchewan flax seeds are rich in Omega-3 fatty acids—lipids essential for building flexible cell membranes.

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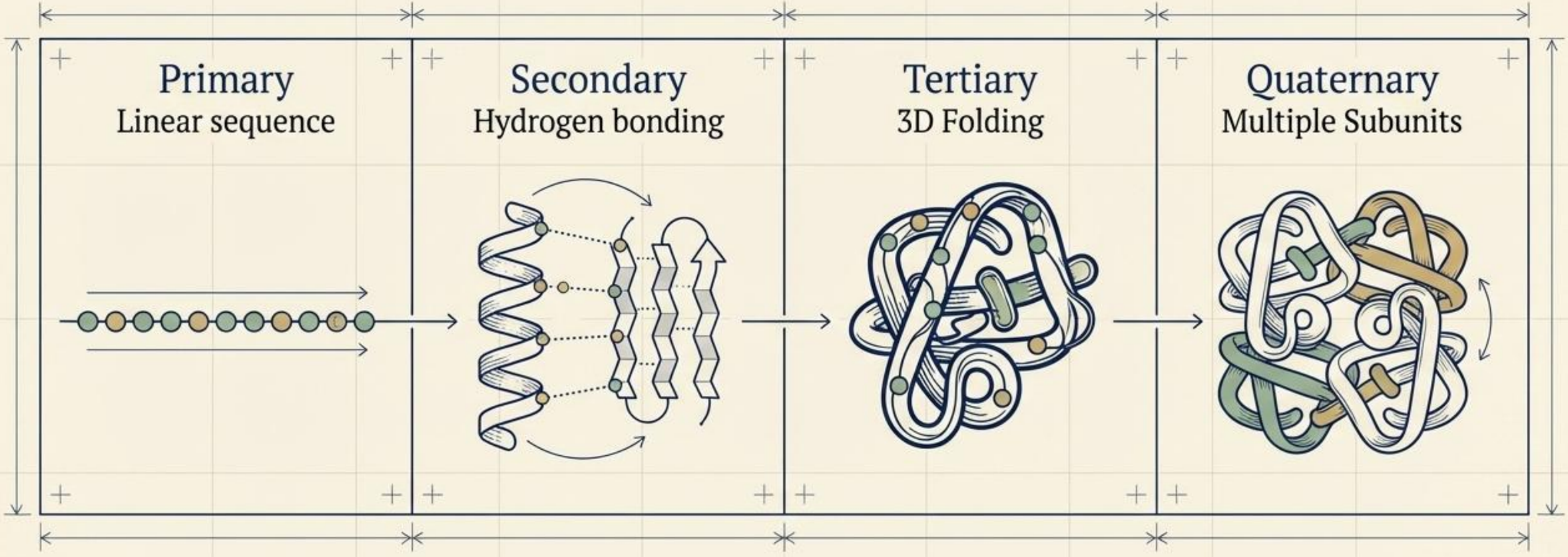
# From String to Machine: Protein Folding

The sequence of amino acids dictates the folding shape, and the shape dictates the protein's function. If hydrogen bonds are disrupted, the 3D structure collapses, destroying the machine.

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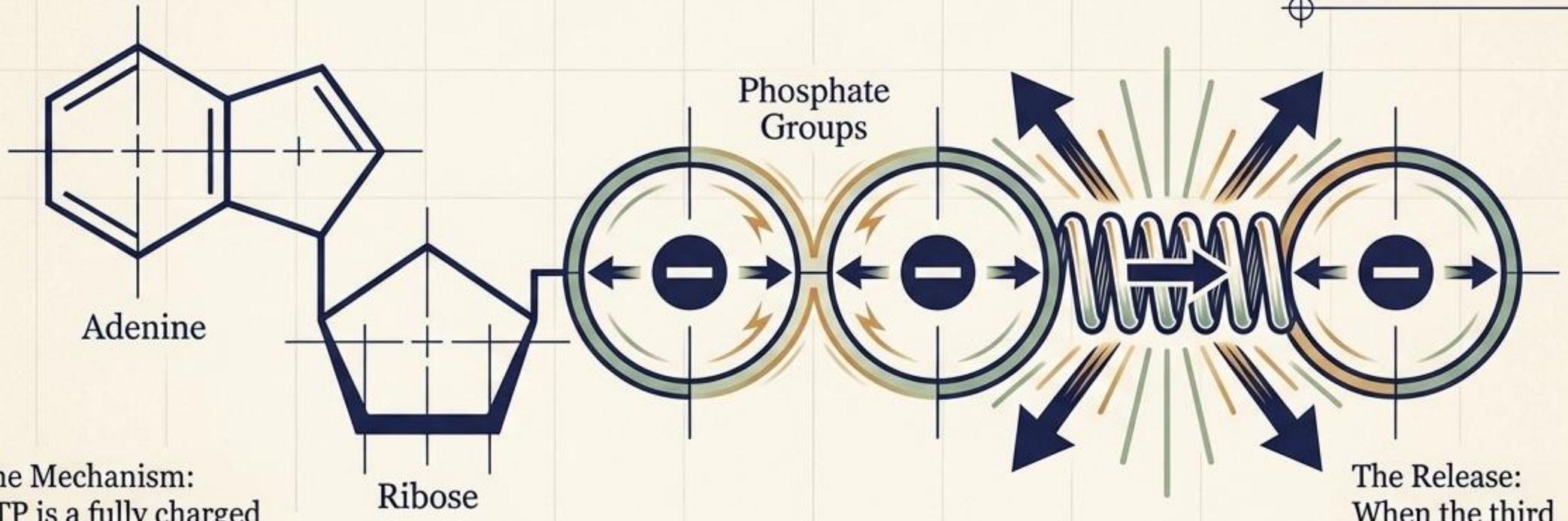
# The Power Currency: Adenosine Triphosphate (ATP)

**Study Prompt:** During your review, identify exactly where in the mitochondria the ADP battery gets recharged.




The Mechanism:
ATP is a fully charged battery. The three
The three phosphate groups are negatively charged and repel each other fiercely.

$$ATP + H_2O \longrightarrow ADP + Pi + Energy$$

(ADP acts as the uncharged battery awaiting recharge)

The Release:
When the third phosphate bond is broken (hydrolysis), the compressed "spring" releases, providing massive cellular energy.

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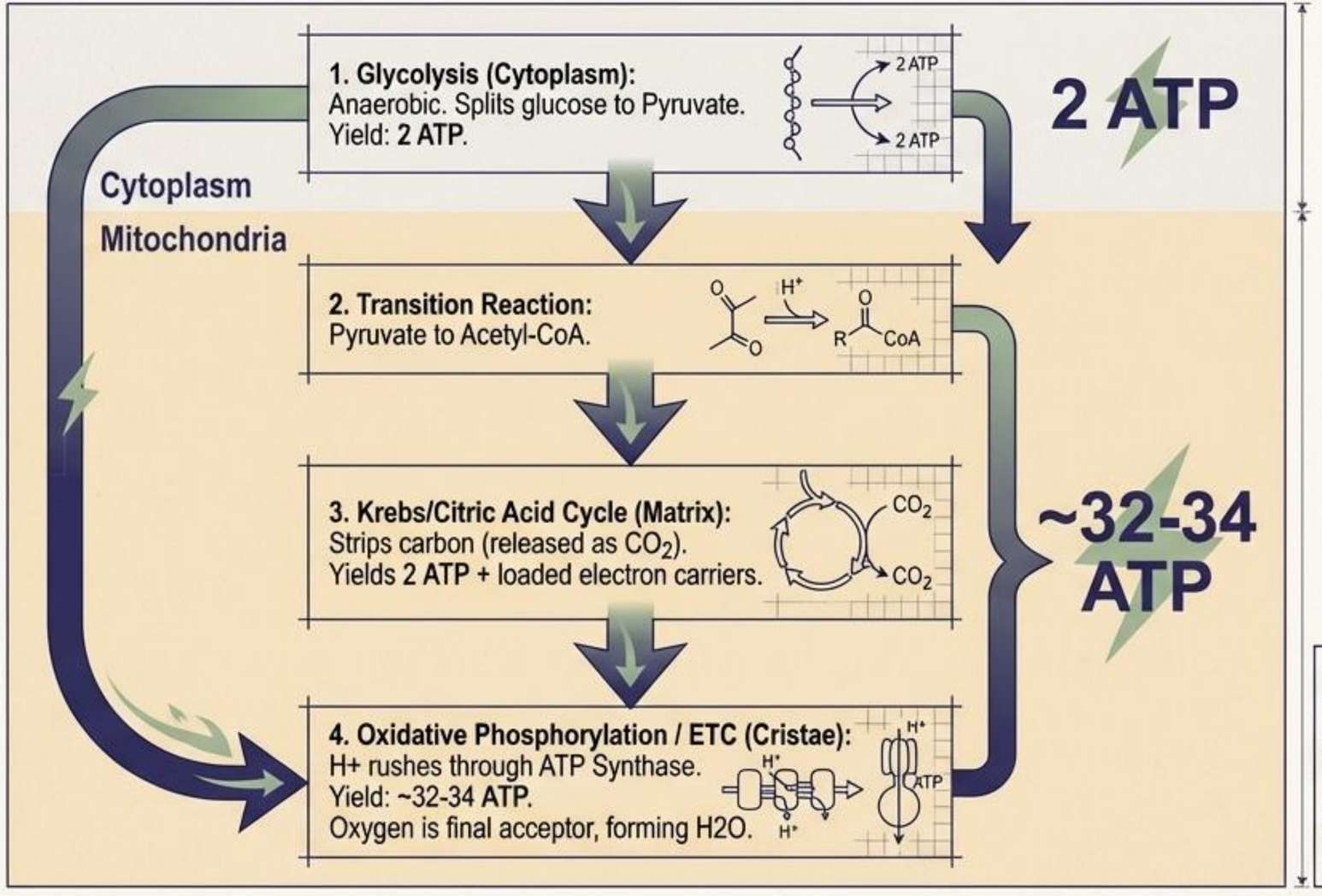
# The Power Grid: Cellular Respiration (The Spender)

**Goal:** Break down Glucose ($C_6H_{12}O_6$) with $O_2$ to yield 36-38 ATP, $CO_2$, and $H_2O$.




**Tracer Callout:**

Scientists tracked radioactive Oxygen (18O) given to cells; it emerges at the end of the ETC as radioactive Water (H2O).

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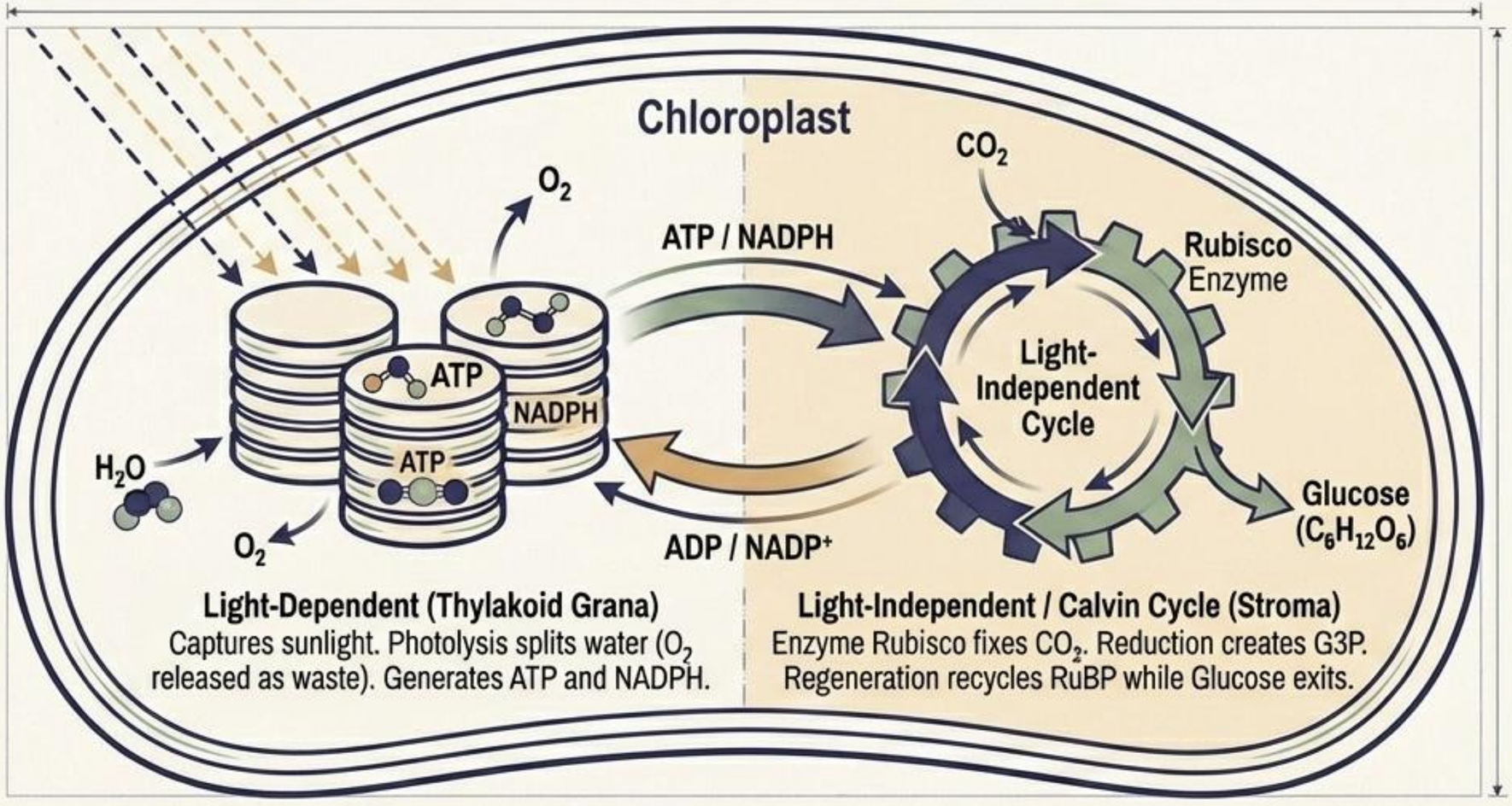
# The Power Grid: Photosynthesis (The Builder)

**Goal:** Use Light + $6CO_2$ + $6H_2O$ to build Glucose ($C_6H_{12}O_6$) and release $6O_2$.




**Context: Solar Efficiency in the Wheat Belt.** U of S researchers study these exact light reactions to maximize starch-rich kernel production during long summer days.

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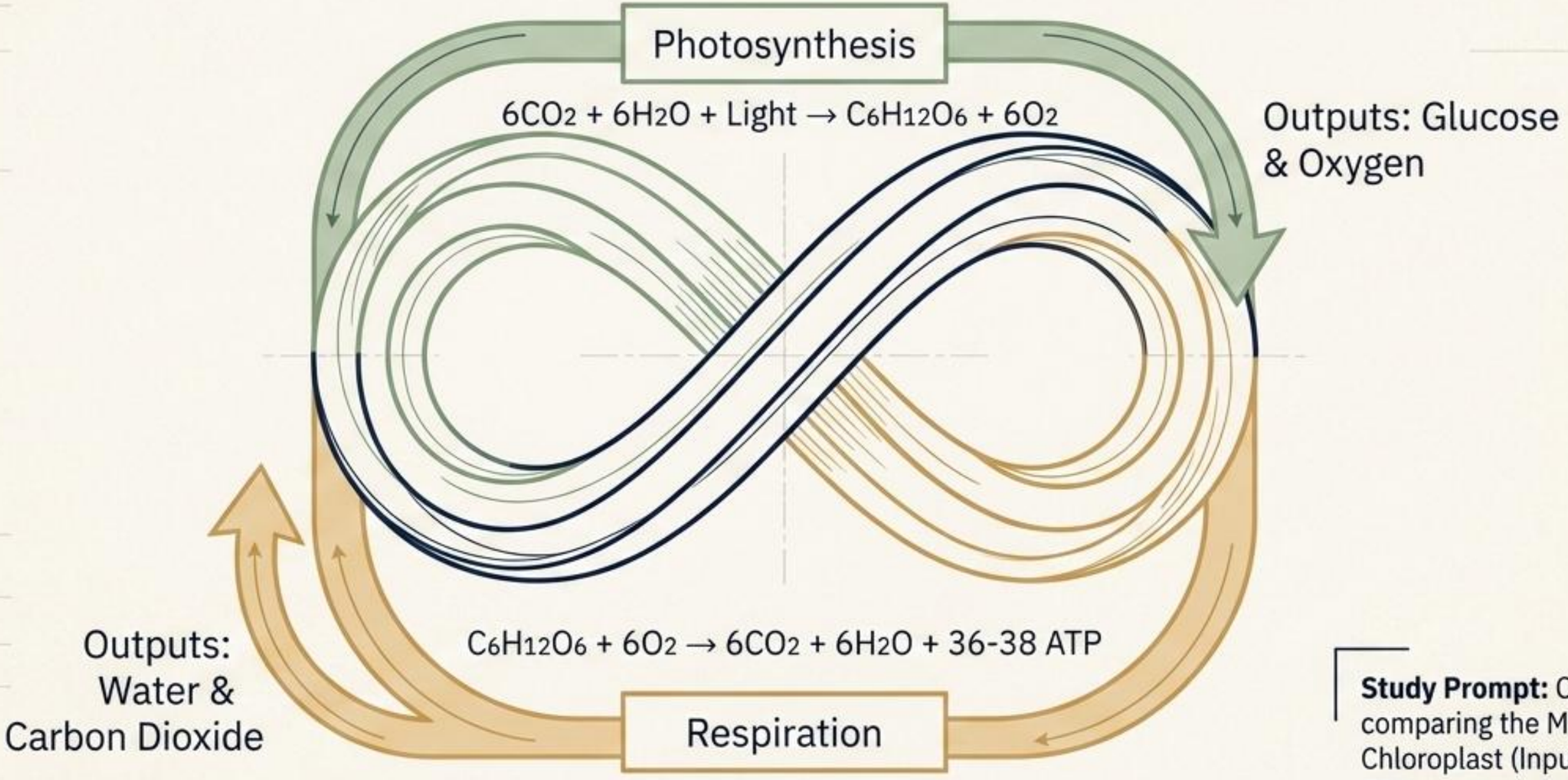
# The Biological Engine: An Inverted Loop

**The Spender vs. The Builder:** The waste products of one system are the vital fuels of the other.




**Study Prompt:** Create a T-Chart comparing the Mitochondrion and Chloroplast (Inputs, Outputs, ETC Location, Cycle Name).

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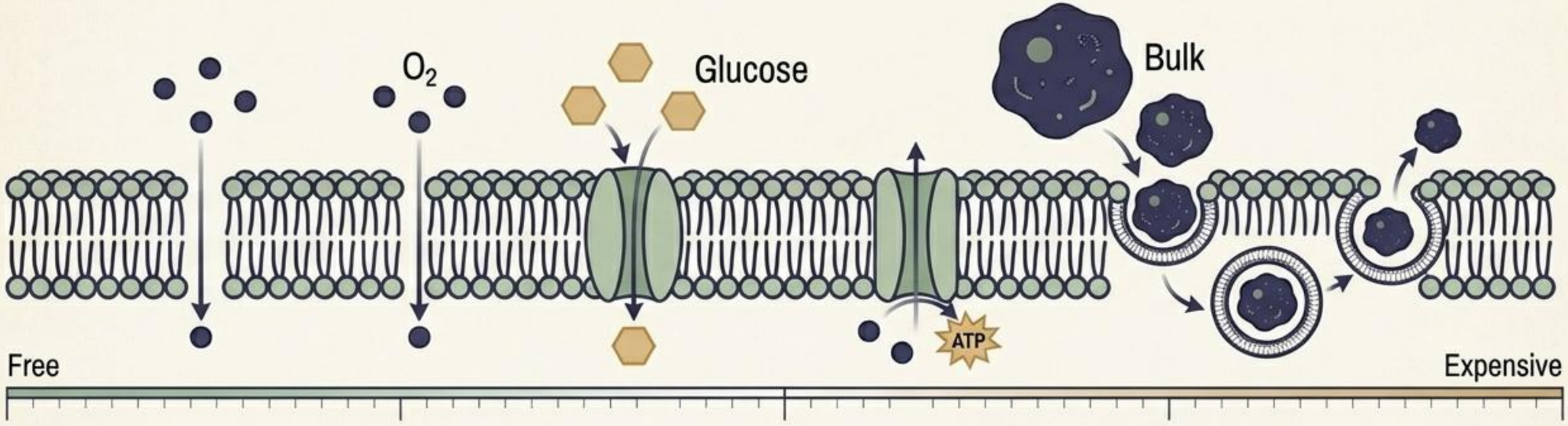
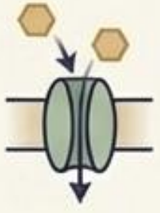
# The Border Patrol: Membrane Transport




## Passive Transport (NO ATP)

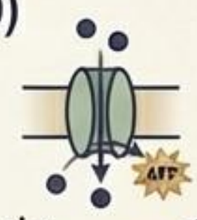
High → Low Gradient

- **Diffusion:** Small particles ($O_2$) pass directly.
- **Osmosis:** Diffusion of water.
- **Facilitated:** Protein channels for larger items.



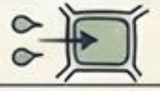
## Active Transport (ATP REQUIRED)

Low → High Gradient




- **Protein Pumps:** E.g., Sodium-Potassium pump.
- **Endocytosis/Exocytosis:** Moving massive bulk via vesicles (Phagocytosis = eating, Pinocytosis = drinking).

**Context: Fertilizer Burn.** Excess nitrogen creates a hypertonic soil environment, drawing water out of plant roots via osmosis.



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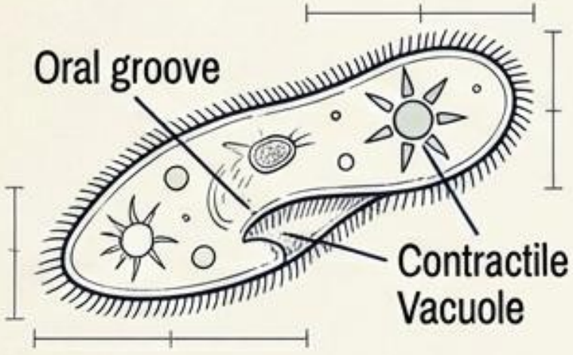
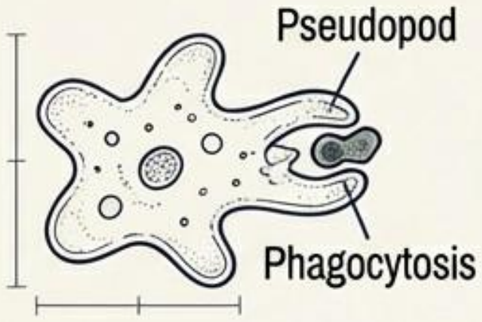
# The Micro-Metropolis & The Extremes

## Unicellular Versatility

- **Paramecium:** Sweeps food via cilia; pumps out excess fresh water via Contractile Vacuole.
- **Amoeba:** Moves and eats (phagocytosis) using Pseudopods.




## Symbiosis Matrix

| | |
|---|---|
| • **Mutualism (+/+):** | Both benefit. |
| • **Commensalism (+/0):** | One benefits, one unaffected. |
| • **Parasitism (+/-):** | One benefits, one harmed. |
| • **Amensalism (-/0):** | One harmed, one unaffected. |

## Life in the Extreme - Archaea

- **Thermophiles** (Heat)
- **Methanogens** (No Oxygen)
- **Halophiles** (Salt lovers—cause the pink water in Saskatchewan's Chaplin Lake).

## Microscopy Tools

- **Light Microscope**: Live cells, ~1000x
- **TEM**: Internal details, dead specimen
- **SEM**: 3D surface mapping, dead specimen.

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